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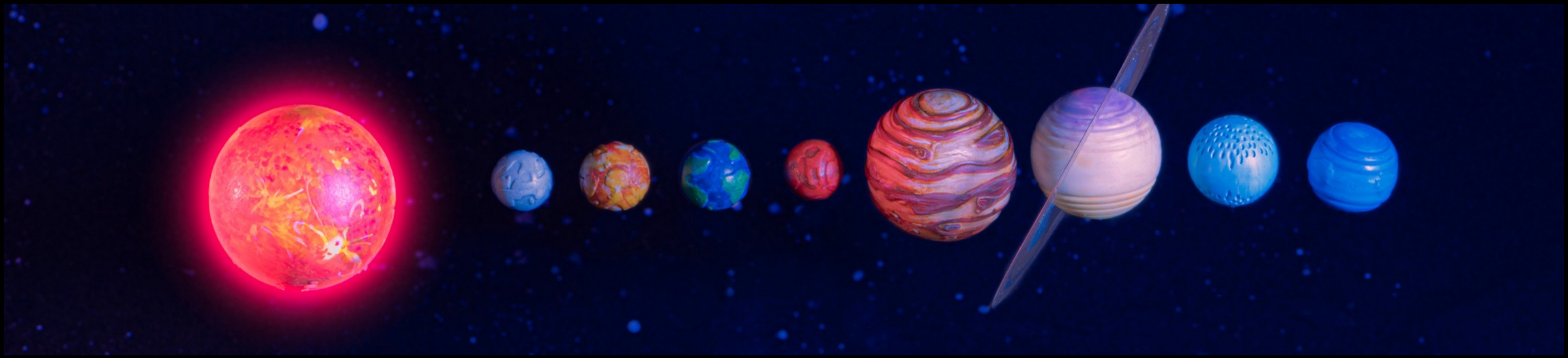
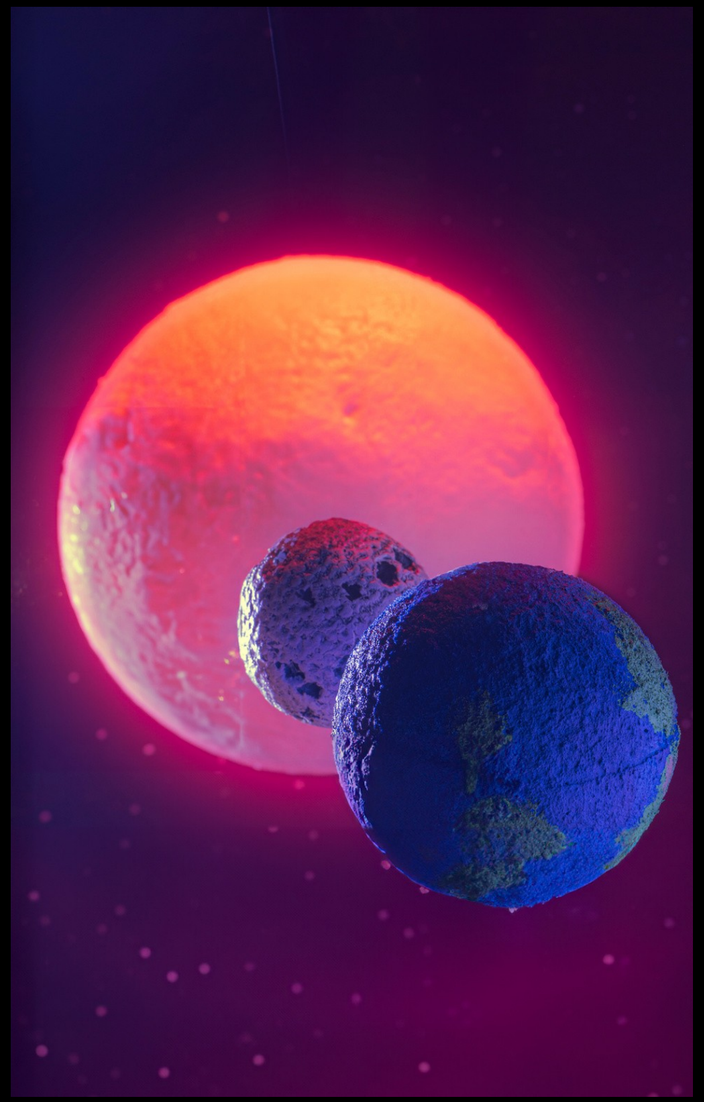


Statistics and Data Analysis methods in Physical Sciences, PHY5132/6132/AOE5132

STATISTICAL ASSESMENT OF POTENTIALLY HABITABLE EXOPLANETS

Submitted By: Abhishank Paul, Anjali C S, Nanda Lakshmi V

MSC25401 MSC25403 MSC25406



CONTENTS

1. INTRODUCTION
2. OBJECTIVE
3. METHOD OF ANALYSIS
4. RESULTS AND DISCUSSION
5. CONCLUSION

1.WHAT ARE EXOPLANETS

- Exoplanets are planets located outside our solar system,orbiting other stars or floating alone as “rogue planets”

- THESE PLANETS ARE CLASSIFIED INTO:
  1. ROCKY PLANETS (EARTH-LIKE)
  2. GAS GIANTS (LIKE JUPITER)
  3. Ice Giants (Neptunian)

1. OBJECTIVE

To analyze exoplanet data and identify Earth-like planets based on relevant planet parameters in order to assess their potential habitability **using statistical and data science methods**.

3.METHOD OF ANALYSIS

1 : DATA COLLECTION

- Exoplanet data was collected from the - Nasa Exoplanet Archive

2 : DATASET PREPROCESSING

- Converting relevant parameters into numerical form
- Removing invalid or missing entries
- Handling of duplicate datas that may have arised due to updation of data set
- Filtering based on error threshold

3 : HABITABILITY TESTING

- Analysis of relevant exoplanet parameters like : Accelration due to gravity, Stellar flux and surface temperature.

METHOD OF ANALYSIS

1 a) Acceleration due to gravity

- AFFECTS ATMOSPHERIC RETENTION, HUMAN SURVIVABILITY, AND OVERALL PLANETARY CONDITIONS.

CODING PART

- Computed surface gravity from mass and radius data in terms of earths gravity.
- Filtered high-error data.

$$\Delta g = \sqrt{(\Delta M)^2 + (2\Delta R)^2}$$

- Fraction of exoplanets having ‘g’ in 0.8g to 1.5g range was computed.
- Correlation between mass, radius and gravity was analyzed using Pearson correlation.
- Visualized the gravity distribution, data reliabitlity vs gravity plot, correlation plot and a tabulated a summary.

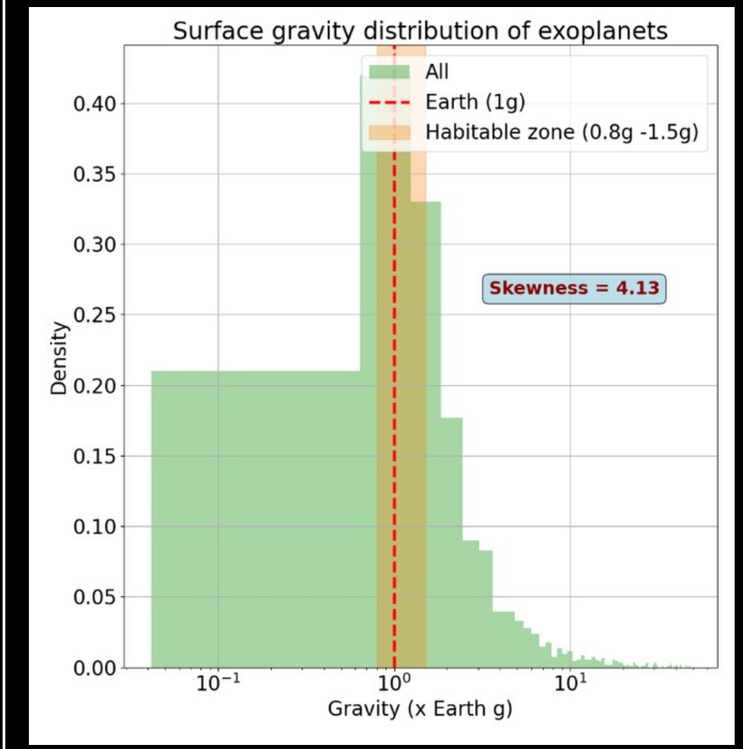
$$g = \frac{M}{R^2}$$

Earth normalized form of g

M = Total mass of the planet
R = Radius of the planet

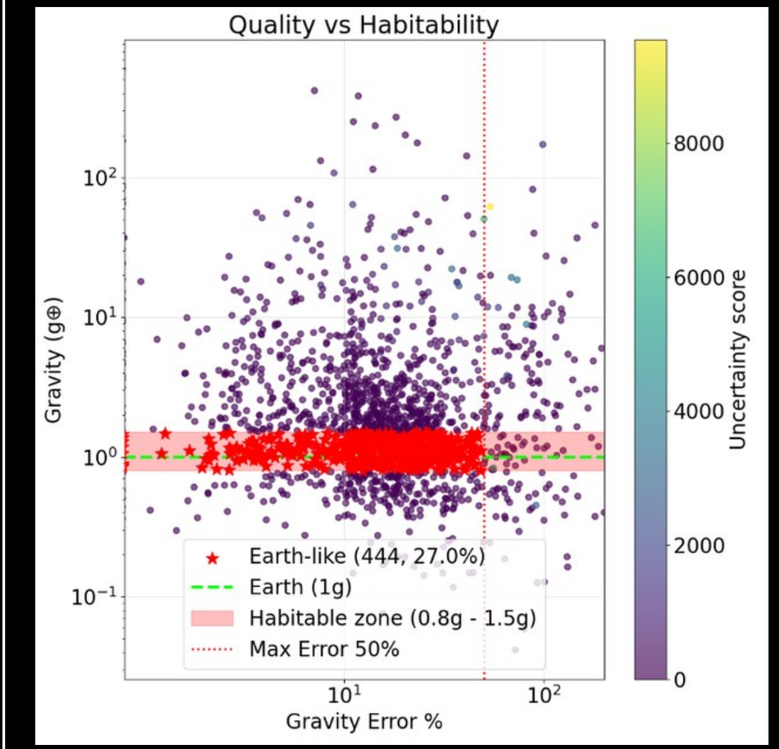
Gravity - Results and Discussion

DENSITY PLOT



- Most planets have low gravity (0.1-1g)
- Extreme right skew (4.13) indicates presence of gas giants
- About 27% of planets lies in the habitable gravity zone

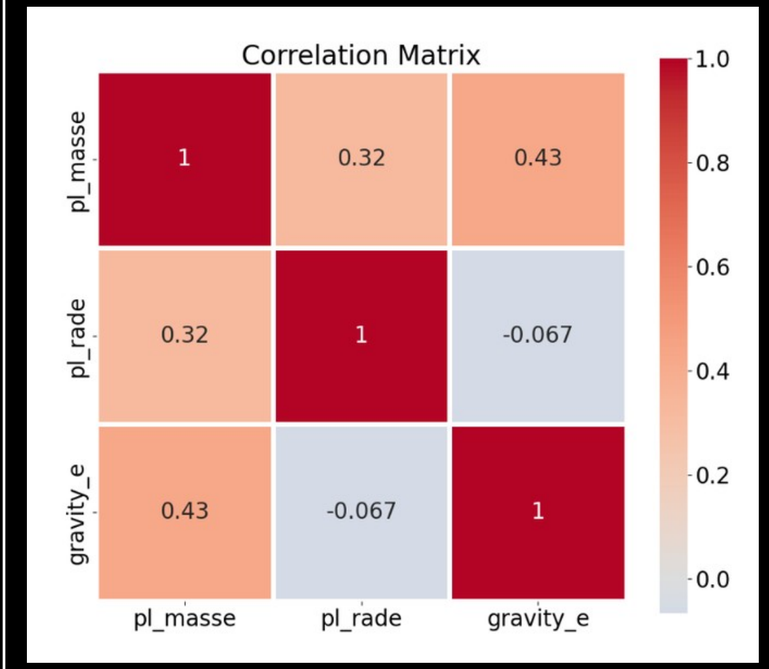
QUALITY PLOT



- Earth-like planets cluster around 1g (27%)
- Most reliable data lies left of error cutoff
- No strong correlation between gravity and error

Gravity - Results and Discussion

CORRELATION PLOT



- Mass shows moderate positive correlation with gravity (0.43)
- Radius shows very weak negative correlation with gravity (-0.067)
- Spearman correlation is more suitable for radius

TABULAR SUMMRY

Parameter	Value
Total Planets	1643
Earth-like Planets	444
Maximum Gravity	597.09
Minimum Gravity	0.042
Mean Gravity	6.29
Median Gravity	1.59
Best Match	TOI-5786 b (g=1.00)

METHOD OF ANALYSIS

3.1 b) Stellar flux

DETERMINES THE ENERGY RECEIVED BY THE PLANETS AND SURFACE TEMPERATURE

CODING PART

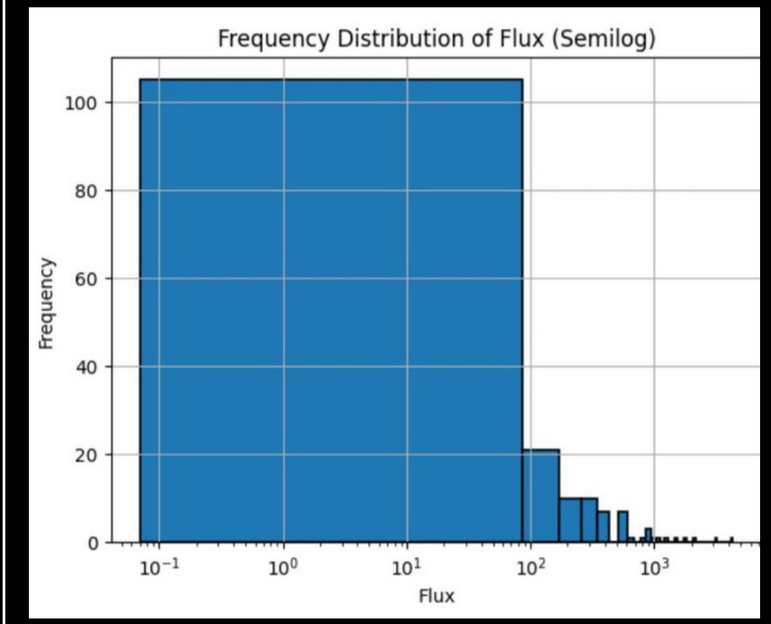
- Computed stellar flux using stellar temperature, radius, and orbital parameters like semi major axis and eccentricity of planets orbit
- Filtered missing values
- Assumed the blackbody radiation model and inverse square law
- Used logarithmic scaling to handle wide data range
- Analyzed flux distribution using histogram
- Studied relationship between flux and uncertainty using scatter plot
- Identified planets within habitable flux range (~0.3 to 1.5 Earth flux)

$$F \propto \frac{R_*^2 T_*^4}{a^2\sqrt{1-e^2}}$$

- T_* = stellar temperature
- R_* = stellar radius
- a = orbital distance
- e = eccentricity

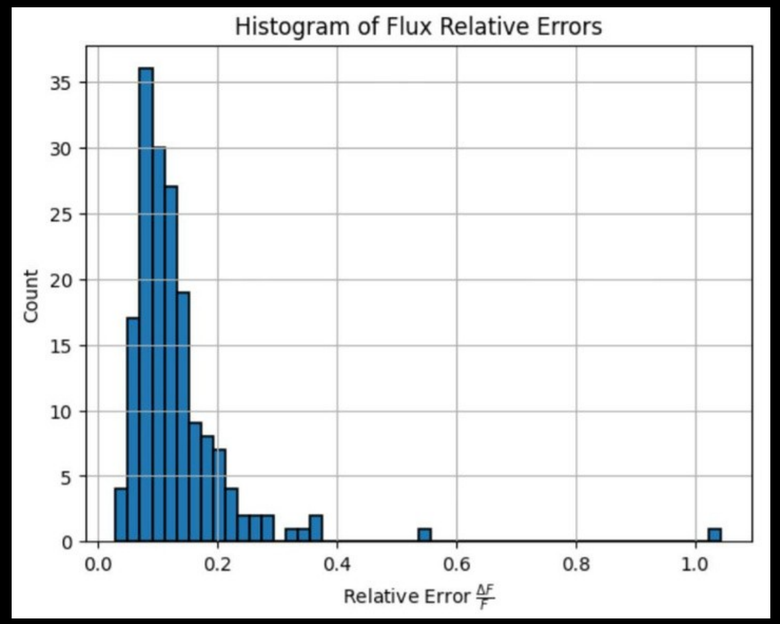
Stellar flux - Results and Discussion

SEMILOG HISTOGRAM



- Highly right skewed
- Majority of exoplanets have low flux values

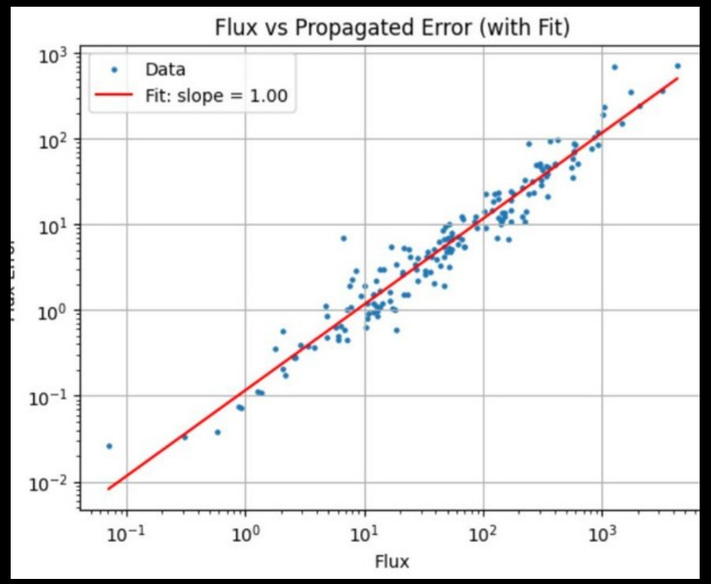
RELATIVE ERROR HISTOGRAM



- Low relative error
- Positively skewed with few high error outliers

Stellar flux - Results and Discussion

SCATTER PLOT



- Strong positive correlation between flux and error
- Error increase linearly with flux

TABULAR SUMMARY

Parameter	Value
Total Planets	6158
Earth-like Planets	0
Maximum Flux	4301.55
Minimum Flux	0.0709
Mean Flux	158.39
Median Flux	36.21
Best Match	Kepler-453 b (Flux: 0.93)

METHOD OF ANALYSIS

3.1 c) Classification of Exoplanets

- EXOPLANETS CAN BE ROCKY, ICE GIANTS, OR GAS GIANTS.TO FIND HABITABLE EXOPLANETS WE NEED FILTER OUT THE ROCKY PLANETS FROM THE DATABASE

CODING PART

- Filtered the planets based on the avilability of the radius R and mass M data
- Used the power law relation between radius and mass
- Utilized the Bayesian approach using Markov Chain Monte Carlo Method.
- Sampled the posterior distribution of (α,k). The distribution chosen was Gaussian
- Accounted for measurement uncertainties
- Explored degeneracies in the mass--radius relation.
- Visualized the categories vs mass and radius using boxplots

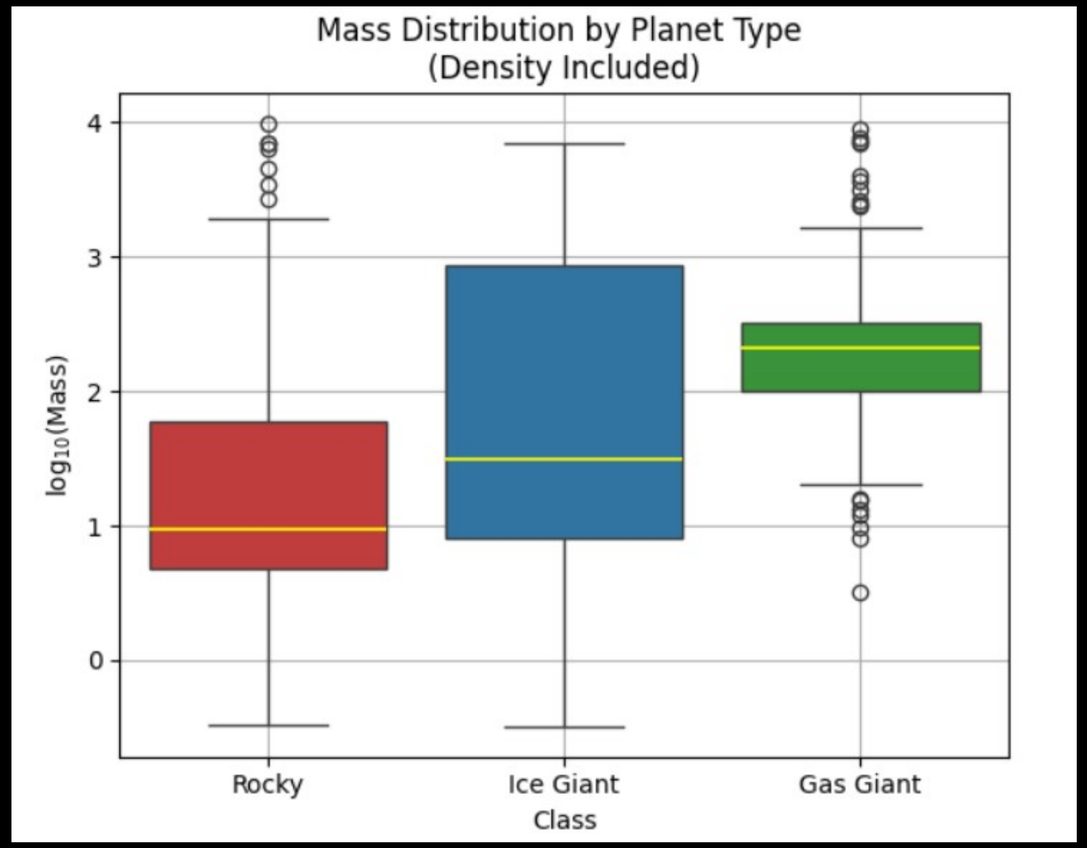
$$R = kM^{\alpha}$$

k = Scaling constant

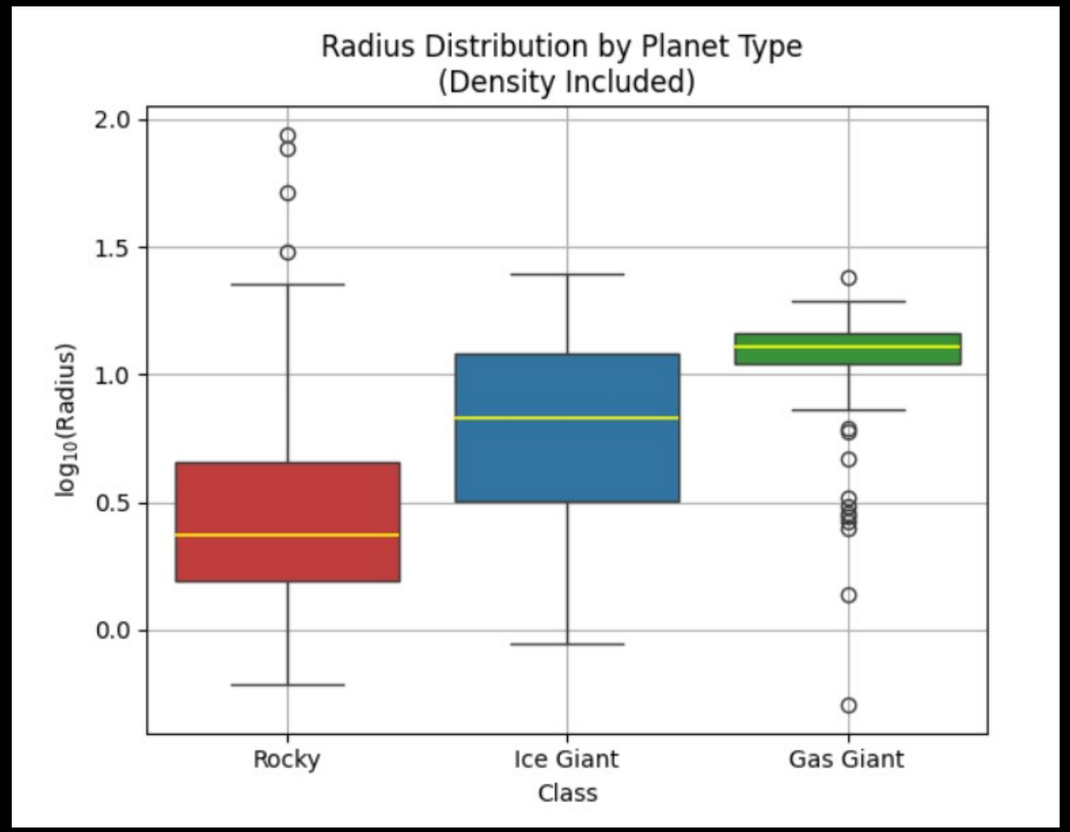
α = Controls how radius scales with mass

M-R CLASSIFICATION - Results

MASS DISTRIBUTION



RADIUS DISTRIBUTION



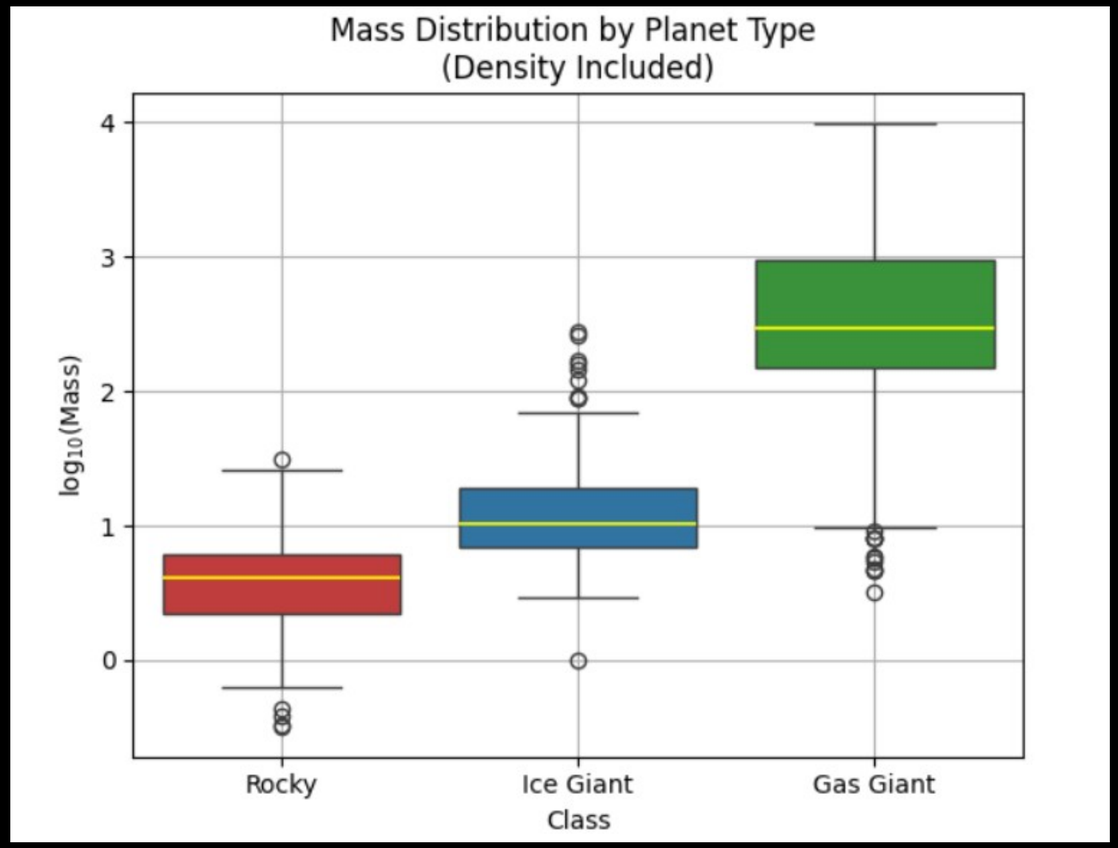
3.1 c) Classification of Exoplanets

- **MASS RADIUS DENSITY CLASSIFCATION**

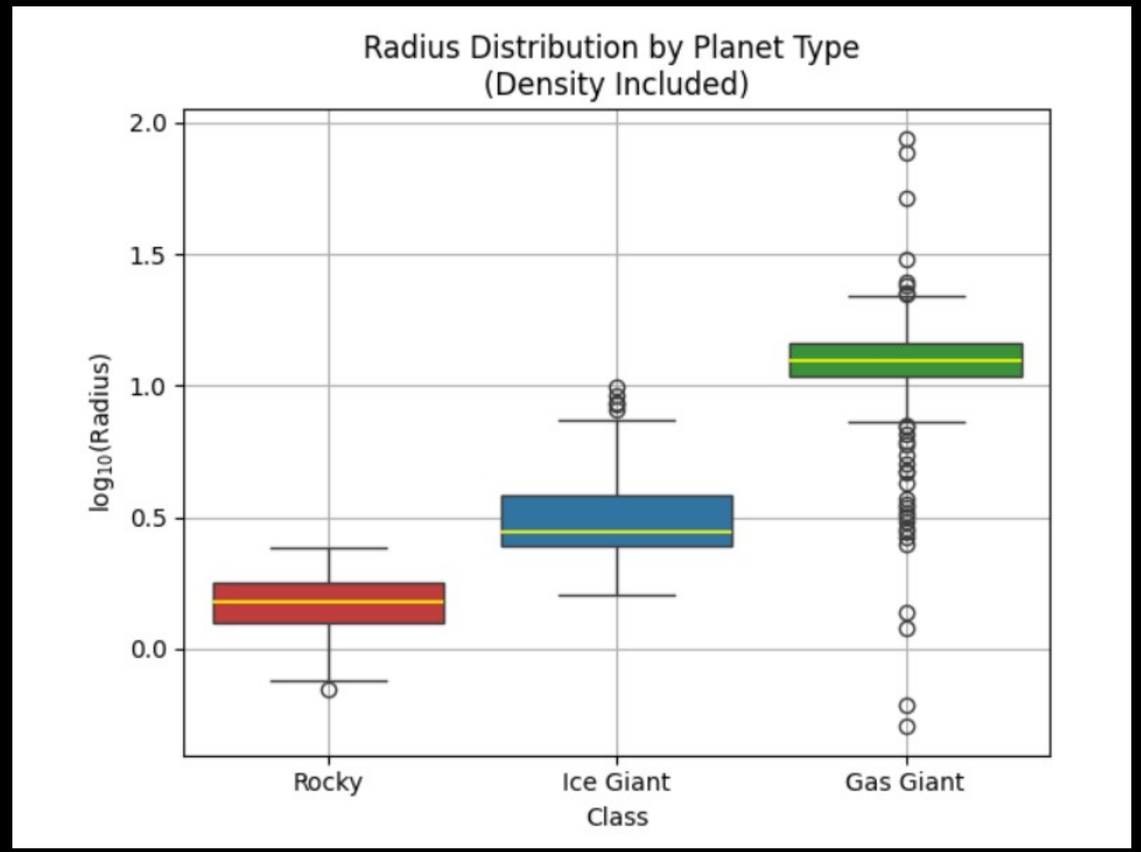
- Mass - radius system is not sufficient because of the mass degeneracy in planet composition
- To resolve this we are using density as a parameter
- Finally we analyzed the correlation between this parameters and found that clssification is much more accurate.
- Still it doesnt account for things like atmospheric stability etc.

M-R-D CLASSIFICATION - Results

MASS DISTRIBUTION

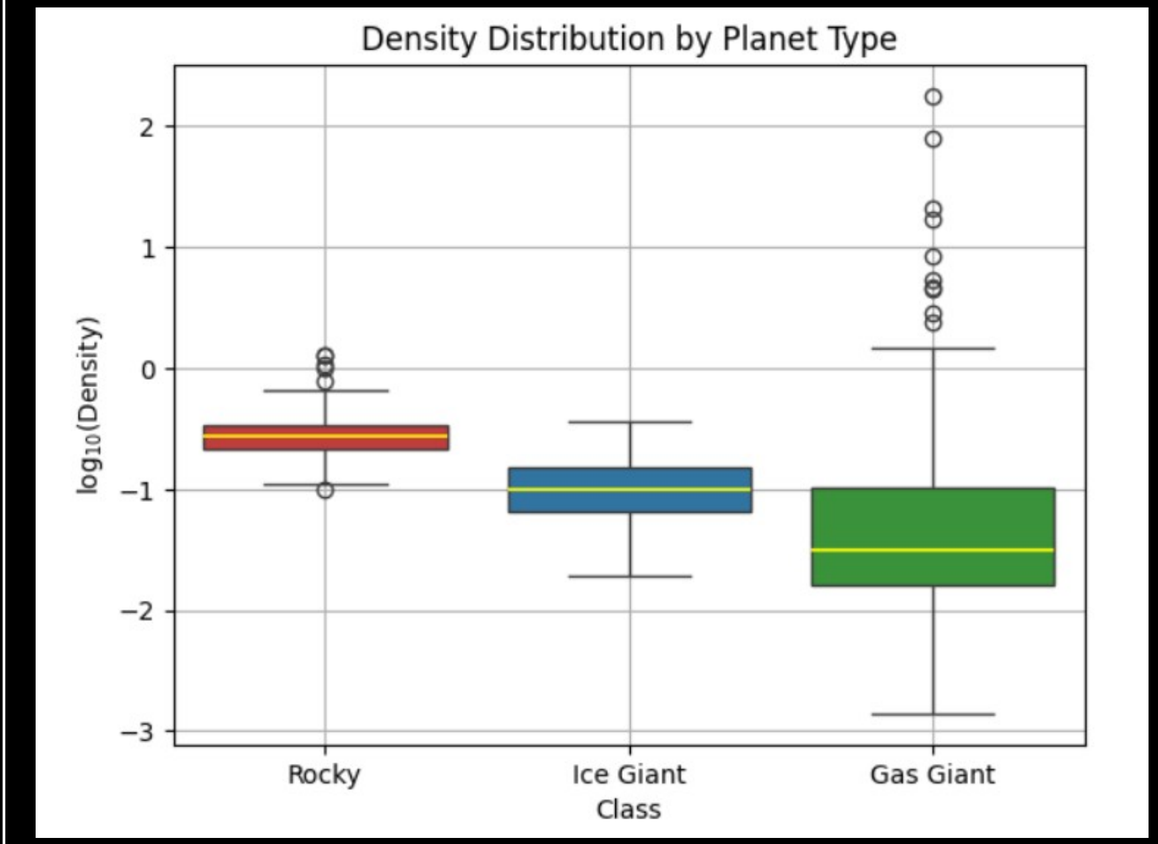


RADIUS DISTRIBUTION



M-R-D CLASSIFICATION - Results

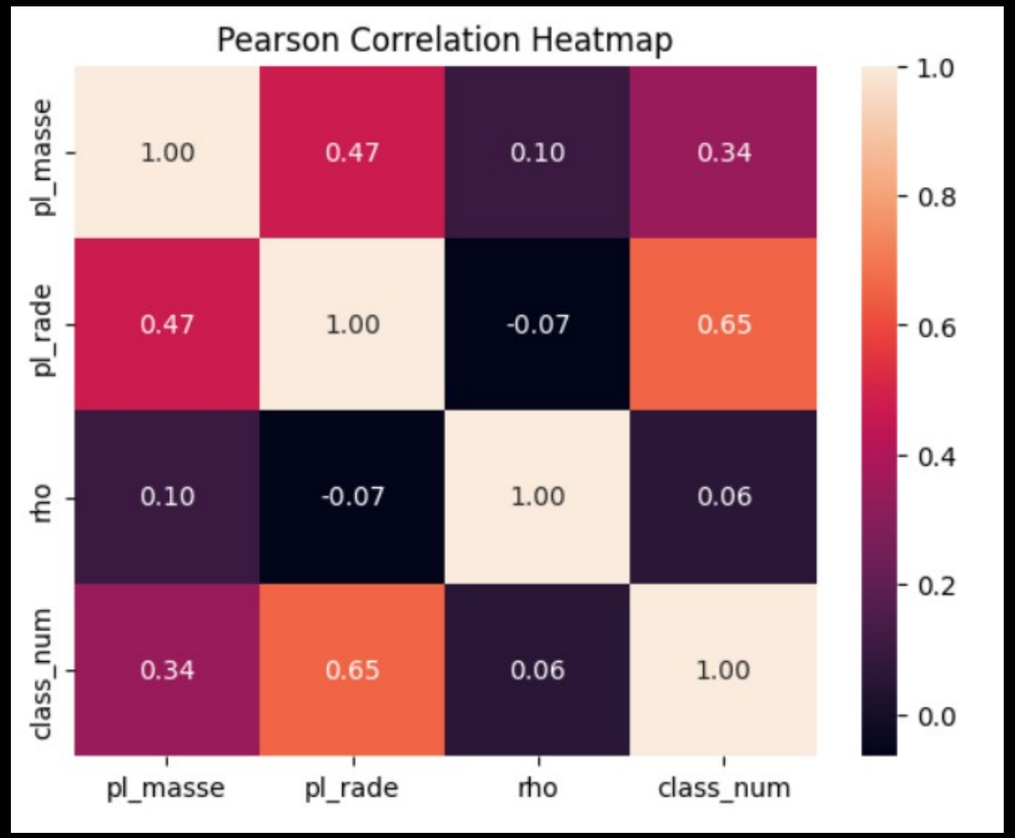
DENSITY DISTRIBUTION



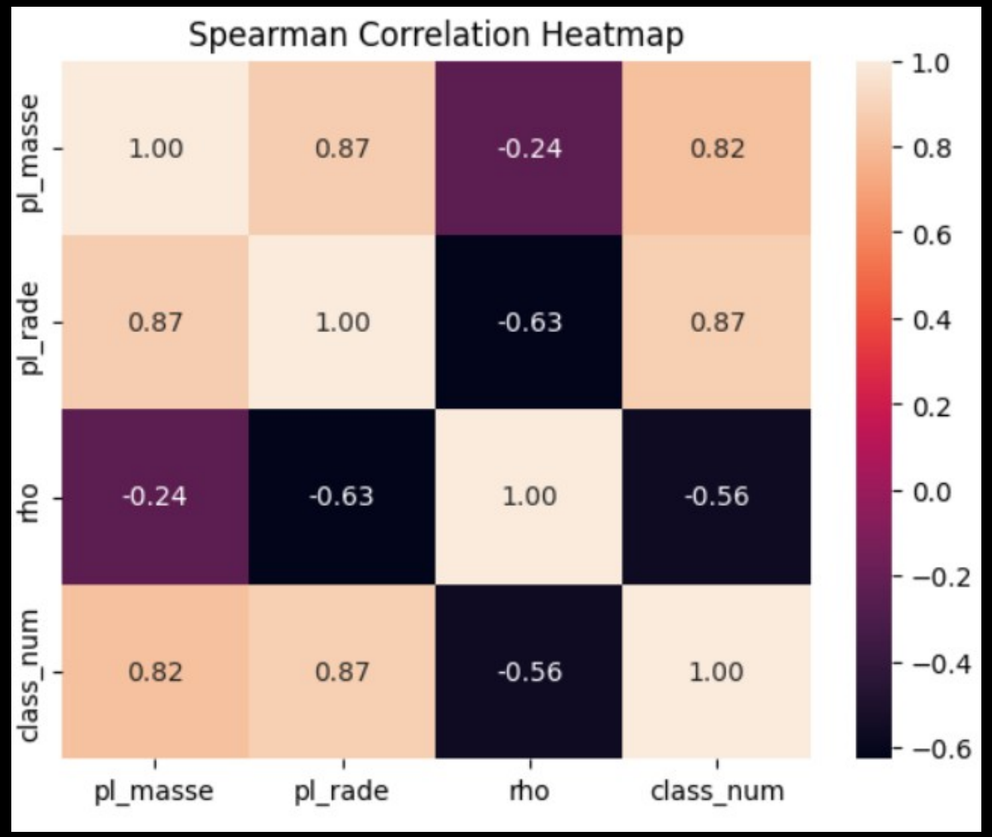
- Rocky and non rocky bodies are clearly seperated indicating density plays the dominant role in classification.
- Ambiguity between warm Neptunes and hot Jupyters still exists.
- Similar issue exist with mini Neptunes and super Earths
- This implies we require atmospheric stability parameters to accurately model.

M-R-D CORRELATION HEAT MAPS

PEARSON HEAT MAP

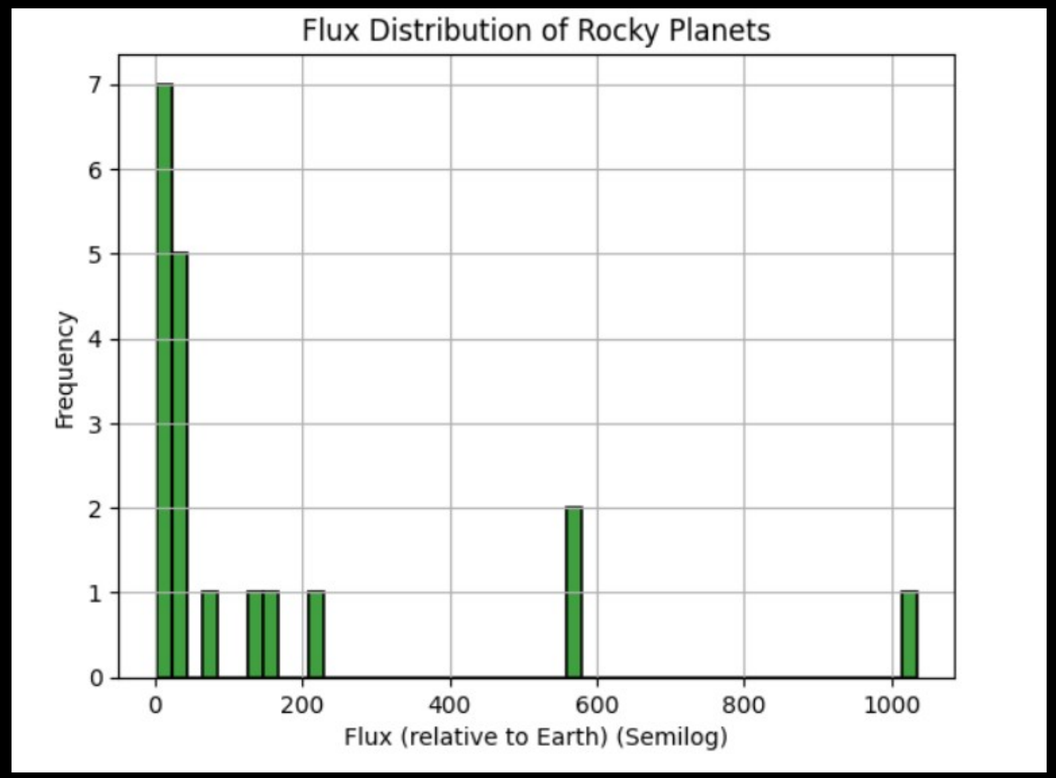


SPEARMAN HEAT MAP

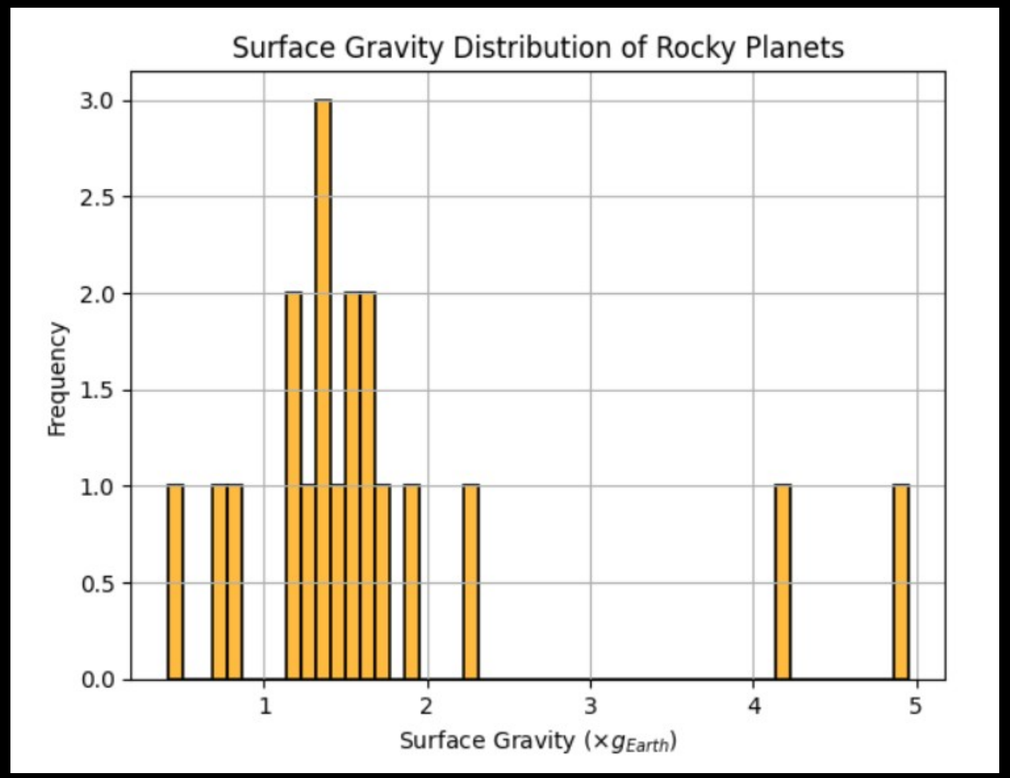


Stellar Flux and Gravity Profile of Rocky Planets

STELLAR FLUX



SURFACE GRAVITY



3.1 d) TEMPERATURE PROFILE

- SURFACE TEMPERATURE IS DETERMINED BY THE STELLAR FLUX , GREENHOUSE EFFECTS AND THE COMPOSITION OF THE ATMOSPHERE

Assumptions

- Grey (Wavelength independent atmosphere)
- Captures greenhouse effect via optical depth

$$\tau = \frac{\kappa P_s}{g}$$

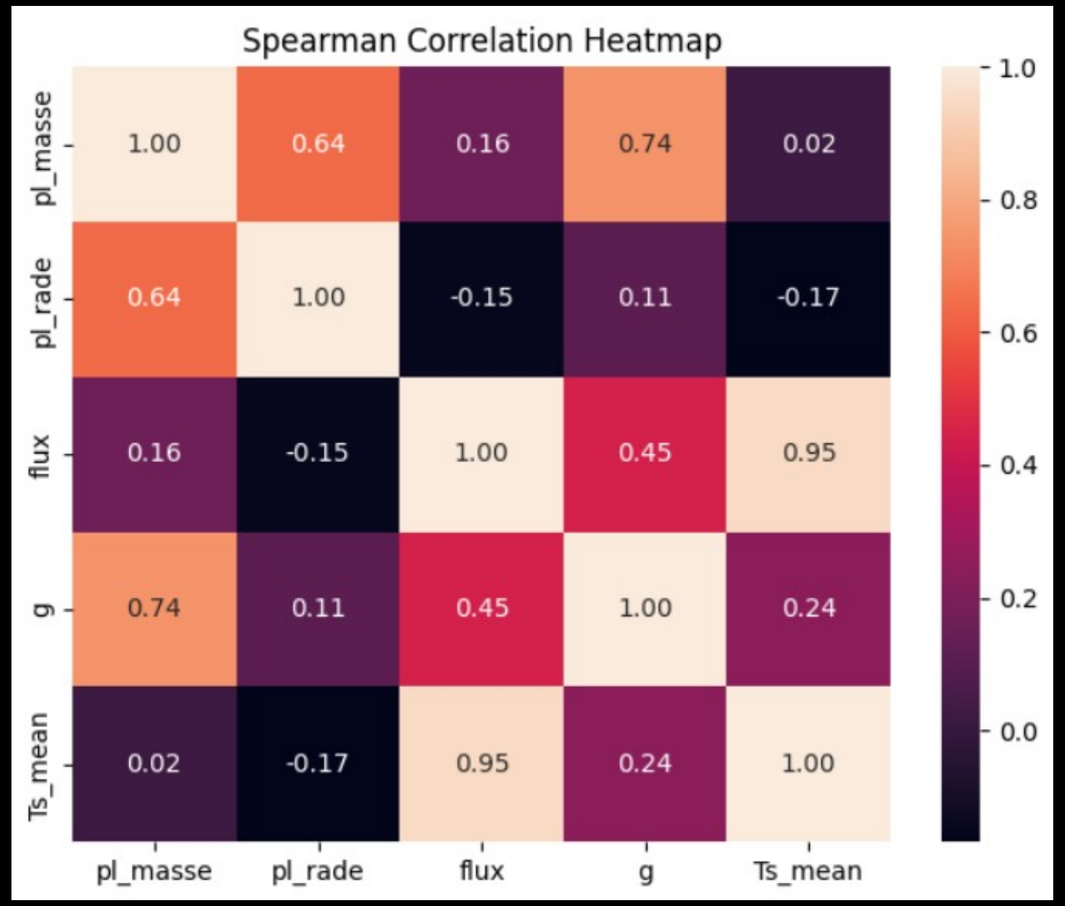
The above term is called optical depth

- This model uses radiative equilibrium and greenhouse effect under hydrostatic equilibrium

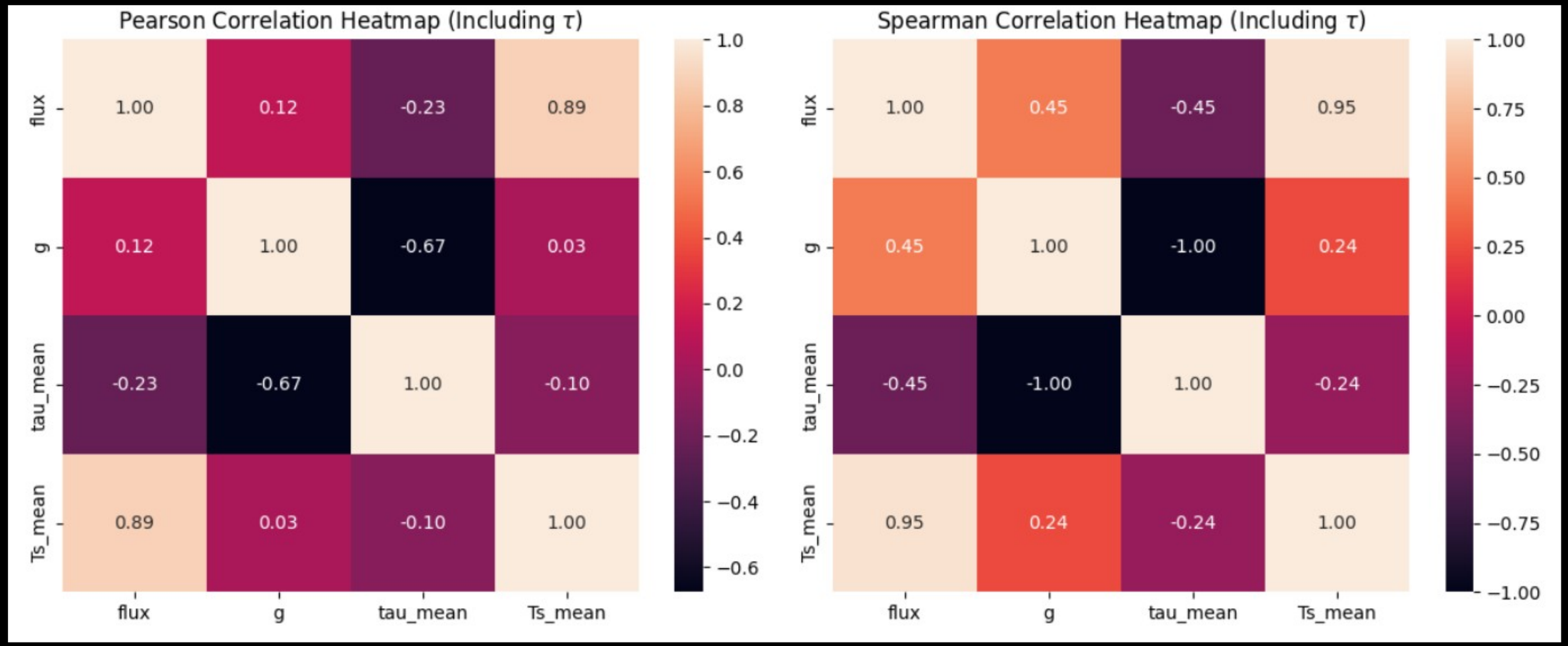
$$T_s = \left(\frac{(1-A)\,F}{4\sigma}\right)^{1/4}\left(1+\frac{3}{4}\frac{\kappa P_s}{g}\right)^{1/4}$$

- A is Albedo,
- F is incident stellar flux
- $\sigma = 5.670374419 \times 10^{-8}$ W/m^2 K^4 is Stefan's constant
- κ is atmospheric opacity
- P_s is surface pressure
- g is acceleration due to gravity

CORRELATION HEAT MAPS



CORRELATION HEAT MAPS



INFERENCE

- Stellar flux is the dominant driver
- Acceleration due to gravity shows negligible inference
- Mass and radius do not directly influence temperature, their effect is only through gravity, so they are redundant variables in the model
- Surface pressure and atmospheric opacity exhibit degeneracy
- The model severely constrains optical depth and the numerator is nearly constant. Only gravity works as a varying parameter.

4. CONCLUSION

- From our model we did not find any exoplanet which may support human life.
- The data set was limited with incomplete data entries which limited our study.
- We only managed to find 19 planets whose complete parameters relevant to our study was available.
- For furthur studies the atmospheric data and composition data is necessary.
- In-situ exploration is required for accurate finding which is still something we need to achieve.

1. $263K\ (-10^{\circ}C) \leq T \leq 333K\ (+60^{\circ}C)$.
2. $0.5 \leq g \leq 2.0$ (In Earth units).
3. $0.5 \leq$ Flux ≤ 5.0 (In Earth Units).

Relaxation criteria for habitability (Contoversial choices indeed)

THANKYOU